

Mohammadreza Nemati

Santa Clara, CA 95054 | Mobile: 567-204-7520 | mnrnemati747@gmail.com | www.linkedin.com/in/mnemati

EXPERIENCE

Senior AI Engineer

05/2025 - Present

Volkswagen Group of America

Belmont, CA

- Optimized Qwen2.5-VL for constrained automotive deployment by depth-pruning 8L, 6L, and 4L student variants, reducing GPU memory footprint while preserving compatible vision-language weights.
- Compressed Qwen2.5-VL 7B into a pruned 1.4B trajectory-prediction model using knowledge distillation and quantization-aware training for lower-memory, deployment-ready inference.
- Built distributed PyTorch/DDP fine-tuning pipelines for multi-task automotive VLM workloads, combining toggle-state prediction, step decomposition, and UI grounding with task-specific evaluation.
- Post-trained multimodal foundation models using SFT, DPO, and GRPO for reasoning, grounding, and task execution in automotive AI workflows.
- Deployed and profiled Qwen3.5 models up to 35B/A3B with TensorRT-LLM on H100/A100 Azure ML endpoints, using NVIDIA Nsight Systems/Compute to diagnose memory versus compute bottlenecks and tune throughput/latency.

Artificial Intelligence Research Assistant

08/2022 - 06/2025

Case Western Reserve University

Cleveland, Ohio

- Optimized LLM inference memory by analyzing KV-cache key/value norm imbalance, achieving 2× memory reduction with >94% accuracy retention across models up to 70B parameters [\[paper link\]](#)
- Developed LIT-LVM using latent variable embeddings, structured regularization, and GNN-based higher-order interactions, outperforming GNN, Factorization Machine, and elastic net baselines in ranking and recommendation tasks [\[paper link\]](#)
- Developed DEASurv, a deep multimodal model leveraging entity-specific embeddings, attention mechanisms, and contrastive loss to improve prediction on large-scale medical datasets

Software Engineer - AI/ML Intern

02/2024 - 07/2024

Sherwin-Williams

Cleveland, Ohio

- Designed and deployed real-time inference pipeline for vision transformers (SAM), optimizing model latency and improving segmentation quality by 7% IoU through inference evaluation and pipeline tuning.
- Built multimodal system integrating LLaVA with vision transformers, implementing cross-modal alignment for natural-language visual grounding and object segmentation.

Data Science Research Intern

05/2023 - 09/2023

Beyond Limits AI

Glendale, California

- Developed and deployed Graph Neural Network models (GCN, GAT, GraphSAGE) using PyTorch Geometric for graph-structured data, achieving 10% accuracy improvement over baseline approaches
- Fine-tuned Llama-2 using PyTorch distributed training with FSDP/DDP for multi-GPU optimization and quantization for efficient deployment.

SKILLS

Programming: Python, C++, CUDA

Performance Optimization: GPU Memory Optimization, Latency/Throughput Tuning, Quantization, Pruning, Knowledge Distillation, Continuous Batching

ML Systems: PyTorch, TensorRT-LLM, Distributed Training (FSDP/DDP), Model Serving, NVIDIA Nsight Systems/Compute, Model Evaluation

Deep Learning: Transformers, Vision Transformers, Multimodal Models, Computer Vision, Reinforcement Learning

EDUCATION

Case Western Reserve University – PhD, Computer Science

Cleveland, OH | 2025

University of Toledo – MS, Computer Science

Toledo, OH | 2020

Shiraz University – BS, Electrical Engineering

Shiraz, Iran | 2017